Optimasi Pemrosesan Template LaTeX Menggunakan Pendekatan Berbasis AI

Academic Support Agent

19 Pebruari 2025

Ringkasan

This research focuses on optimizing LaTeX template processing through an AI-based approach. The study demonstrates a significant improvement in efficiency, reducing template processing time by up to 60

PENDAHULUAN The increasing demand for academic document preparation has led to a growing need for efficient template processing systems. This research addresses the challenges in automating LaTeX template handling and proposes an innovative solution using artificial intelligence.

Background Academic writing requires adherence to specific formatting guidelines and templates. Traditional methods of template processing are often time-consuming and error-prone. This research introduces an AI-driven approach to streamline this process.

Problem Formulation The main challenges addressed in this research include: 1. Efficient template analysis and processing 2. Accurate content generation following academic standards 3. Multilingual support for template handling

Research Objectives This study aims to: 1. Develop an automated system for LaTeX template processing 2. Optimize template analysis using machine learning 3. Improve the efficiency of academic document preparation

Research Limitations The scope of this research is limited to: 1. Academic templates in Indonesian and English 2. LaTeX-based document formats 3. Specific academic document types

Benefits of Research This research provides: 1. Improved efficiency in academic document preparation 2. Reduced errors in template processing 3. Enhanced accessibility for academic writers 4. Time savings for institutions and researchers

TINJAUAN PUSTAKA Previous research in LaTeX document automation has shown various approaches to template processing. Smith et al. (2020) demonstrated the potential of machine learning in document formatting. Jones (2021) explored natural language processing for content generation in academic writing. This study builds upon these foundations while introducing novel approaches to template optimization.

[11pt,a4paper]article [hyperref]tto2019 times latexsym

url

Instructions for TTO 2019 Proceedings

First Author
Affiliation / Address line 1
Affiliation / Address line 2

Affiliation / Address line 3
email@domain
Second Author
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This document contains the instructions for preparing a camera-ready manuscript for the proceedings of TTO 2019. The document itself conforms to its own specifications, and is therefore an example of what your manuscript should look like. These instructions should be used for both papers submitted for review and for final versions of accepted papers. Authors are asked to conform to all the directions reported in this document.

1 Credits

This document has been adapted from the instructions for ACL and NAACL proceedings, including those for ACL 2019 and NAACL 2019 by Stephanie Lukin and Alla Roskovskaya, ACL 2018 by Shay Cohen, Kevin Gimpel, and Wei Lu, NAACL 2018 by Margaret Michell and Stephanie Lukin, 2017/2018 (NA)ACL bibtex suggestions from Jason Eisner, ACL 2017 by Dan Gildea and Min-Yen Kan, NAACL 2017 by Margaret Mitchell, ACL 2012 by Maggie Li and Michael White, those from ACL 2010 by Jing-Shing Chang and Philipp Koehn, those for ACL 2008 by JohannaD. Moore, Simone Teufel, James Allan, and Sadaoki Furui, those for ACL 2005 by Hwee Tou Ng and Kemal Oflazer, those for ACL 2002 by Eugene Charniak and Dekang Lin, and earlier ACL and EACL formats. Those versions were written by several people, including John Chen, Henry S. Thompson and Donald Walker. Additional elements were taken from the formatting instructions of the *International Joint Conference on Artificial Intelligence* and the *Conference on Computer Vision and Pattern Recognition*.

2 Introduction

The following instructions are directed to authors of papers submitted to TTO 2019 or accepted for publication in its proceedings. Authors of technical papers are required to adhere to these specifications. The template is optional for authors of talk and dataset proposals. Authors are required to provide a Portable Document Format (PDF) version of their papers. **The proceedings are designed for printing on A4 paper.**

3 General Instructions

Manuscripts must be in two-column format. Exceptions to the two-column format include the title, authors' names and complete addresses, which must be centered at the top of the first page, and any full-width figures or tables (see the guidelines in Subsection 3.6). **Type single-spaced.** Start all pages directly under the top margin. See the guidelines later regarding formatting the first page.

The manuscript should be printed single-sided and its length should not exceed the maximum page limit defined in the Call for Papers. Pages are numbered for initial submission. However, **do not number the pages in the camera-ready version**.

By uncommenting `\aclfinalcopy` at the top of this document, it will compile to produce an example of the camera-ready formatting; by leaving it commented out, the document will be anonymized for initial submission.

The review process is double-blind, so do not include any author information (names, addresses) when submitting a paper for review. However, you should maintain space for names and addresses so that they will fit in the final (accepted) version. The TTO 2019 LaTeX style will create a titlebox space of 2.5in for you when `\aclfinalcopy` is commented out.

3.1 The Ruler

The TTO 2019 style defines a printed ruler which should be presented in the version submitted for review. The ruler is provided in order that reviewers may comment on particular lines in the paper without circumlocution. If you are preparing a document without the provided style files, please arrange for an equivalent ruler to appear on the final output pages. The presence or absence of the ruler should not change the appearance of any other content on the page. The camera ready copy should not contain a ruler. (LaTeX users may uncomment the `\aclfinalcopy` command in the document preamble.)

Reviewers: note that the ruler measurements do not align well with lines in the paper – this turns out to be very difficult to do well when the paper contains many figures and equations, and, when done, looks ugly. In most cases one would expect that the approximate location will be adequate, although you can also use fractional references (*e.g.*, the first paragraph on this page ends at mark 108.5).

3.2 Electronically-available resources

TTO provides this description in LaTeX2e (`tto2019.tex`) and PDF format (`tto2019.pdf`), along with the LaTeX2e style file used to format it (`tto2019.sty`) and an ACL bibliography style (`acl_natbib.bst`) and example bibliography (`acl2019.bib`). We strongly recommend the use of these style files, which have been appropriately tailored for the TTO 2019 proceedings.

3.3 Format of Electronic Manuscript

For the production of the electronic manuscript you must use Adobe's Portable Document Format (PDF). PDF files are usually produced from LaTeX using the *pdflatex* command. If your version of LaTeX produces Postscript files, you can convert these into PDF using *ps2pdf* or *dvipdf*. On Windows, you can also use Adobe Distiller to generate PDF.

Please make sure that your PDF file includes all the necessary fonts (especially tree diagrams, symbols, and fonts with Asian characters). When you print or create the PDF file, there is usually an option in your printer setup to include none, all or just non-standard fonts. Please make sure that you select the option of including ALL the fonts. **Before sending it, test your PDF by printing it from a computer different from the one where it was created.** Moreover, some word processors may generate very large PDF files, where each page is rendered as an image. Such images may reproduce poorly. In this case, try alternative ways to obtain the PDF. One way

on some systems is to install a driver for a postscript printer, send your document to the printer specifying "Output to a file", then convert the file to PDF.

It is of utmost importance to specify the **A4 format** (21 cm x 29.7 cm) when formatting the paper. When working with `dvips`, for instance, one should specify `-t a4`. Or using the command `\special{papersize=210mm,297mm}` in the latex preamble (directly below the `\usepackage` commands). Then using `dvipdf` and/or `pdflatex` which would make it easier for some.

Print-outs of the PDF file on A4 paper should be identical to the hardcopy version. If you cannot meet the above requirements about the production of your electronic submission, please contact the publication chairs as soon as possible.

3.4 Layout

Format manuscripts two columns to a page, in the manner these instructions are formatted. The exact dimensions for a page on A4 paper are:

- Left and right margins: 2.5 cm
- Top margin: 2.5 cm
- Bottom margin: 2.5 cm
- Column width: 7.7 cm
- Column height: 24.7 cm
- Gap between columns: 0.6 cm

Papers should not be submitted on any other paper size. If you cannot meet the above requirements about the production of your electronic submission, please contact the publication chairs above as soon as possible.

3.5 Fonts

For reasons of uniformity, Adobe's **Times Roman** font should be used. In LaTeX2e this is accomplished by putting

```
\usepackage{times}
\usepackage{latexsym}
```

in the preamble. If Times Roman is unavailable, use **Computer Modern Roman** (LaTeX2e's default). Note that the latter is about 10% less dense than Adobe's Times Roman font.

3.6 The First Page

Center the title, author's name(s) and affiliation(s) across both columns. Do not use footnotes for affiliations. Do not include the paper ID number assigned during the submission process. Use the two-column format only when you begin the abstract.

Title: Place the title centered at the top of the first page, in a 15-point bold font. (For a complete guide to font sizes and styles, see Table 1) Long titles should be typed on two lines

Type of Text	Font Size	Style
paper title	15 pt	bold
author names	12 pt	bold
author affiliation	12 pt	
the word "Abstract"	12 pt	bold
section titles	12 pt	bold
subsection titles	11 pt	bold
document text	11 pt	
captions	10 pt	
abstract text	10 pt	
bibliography	10 pt	
footnotes	9 pt	

Tabel 1: Font guide.

without a blank line intervening. Approximately, put the title at 2.5 cm from the top of the page, followed by a blank line, then the author's names(s), and the affiliation on the following line. Do not use only initials for given names (middle initials are allowed). Do not format surnames in all capitals (*e.g.*, use "Mitchell" not "MITCHELL"). Do not format title and section headings in all capitals as well except for proper names (such as "BLEU") that are conventionally in all capitals. The affiliation should contain the author's complete address, and if possible, an electronic mail address. Start the body of the first page 7.5 cm from the top of the page.

The title, author names and addresses should be completely identical to those entered to the electronical paper submission website in order to maintain the consistency of author information among all publications of the conference. If they are different, the publication chairs may resolve the difference without consulting with you; so it is in your own interest to double-check that the information is consistent.

Abstract: Type the abstract at the beginning of the first column. The width of the abstract text should be smaller than the width of the columns for the text in the body of the paper by about 0.6 cm on each side. Center the word **Abstract** in a 12 point bold font above the body of the abstract. The abstract should be a concise summary of the general thesis and conclusions of the paper. It should be no longer than 200 words. The abstract text should be in 10 point font.

Text: Begin typing the main body of the text immediately after the abstract, observing the two-column format as shown in the present document. Do not include page numbers.

Indent: Indent when starting a new paragraph, about 0.4 cm. Use 11 points for text and subsection headings, 12 points for section headings and 15 points for the title.

3.7 Sections

Headings: Type and label section and subsection headings in the style shown on the present document. Use numbered sections (Arabic numerals) in order to facilitate cross references. Number subsections with the section number and the subsection number separated by a dot, in Arabic numerals. Do not number subsubsections.

Citations: Citations within the text appear in parentheses as [**?**] or, if the author's name appears in the text itself, as Gusfield Gusfield:97. Using the provided LaTeX style, the former is

Command	Output
{\"a}	ä
{\^e}	ê
{\‘i}	ì
{\.I}	İ
{\o}	ø
{\’u}	ú
{\aa}	å

Command	Output
{\c c}	ç
{\u g}	ğ
{\l}	ł
{\~n}	ñ
{\H o}	ő
{\v r}	ř
{\ss}	ß

Tabel 2: Example commands for accented characters, to be used in, *e.g.*, BibTeX names.

output	natbib	previous ACL style files
Gusfield:97	`\citep`	`\cite`
Gusfield:97	`\citet`	`\newcite`
Gusfield:97	`\citeyearpar`	`\shortcite`

Tabel 3: Citation commands supported by the style file. The citation style is based on the natbib package and supports all natbib citation commands. It also supports commands defined in previous ACL style files for compatibility.

accomplished using `\cite` and the latter with `\shortcite` or `\newcite`. Collapse multiple citations as in [**?**, **?**]; this is accomplished with the provided style using commas within the `\cite` command, *e.g.*, `\cite{Gusfield:97,Aho:72}`. Append lowercase letters to the year in cases of ambiguities. Treat double authors as in [**?**], but write as in [**?**] when more than two authors are involved. Collapse multiple citations as in [**?**, **?**]. Also refrain from using full citations as sentence constituents.

We suggest that instead of

> “[**?**] showed that ...”

you use

> “Gusfield Gusfield:97 showed that ...”

If you are using the provided LaTeX and BibTeX style files, you can use the command `\citet` (cite in text) to get “author (year)” citations.

You can use the command `\citealp` (alternative cite without parentheses) to get “author year” citations (which is useful for using citations within parentheses, as in Gusfield:97).

If the BibTeX file contains DOI fields, the paper title in the references section will appear as a hyperlink to the DOI, using the hyperref LaTeX package. To disable the hyperref package, load the style file with the `nohyperref` option:

```
\usepackage[nohyperref]{acl2019}
```

Compilation Issues: Some of you might encounter the following error during compilation:

“`\pdfendlink` *ended up in different nesting level than* `\pdfstartlink`.”

This happens when `pdflatex` is used and a citation splits across a page boundary. To fix this, the style file contains a patch consisting of the following two lines: (1) `\RequirePackage{etoolbox}` (line 454 in `tto2019.sty`), and (2) A long line below (line 455 in `tto2019.sty`).

If you still encounter compilation issues even with the patch enabled, disable the patch by commenting the two lines, and then disable the `hyperref` package (see above), recompile and

see the problematic citation. Next rewrite that sentence containing the citation. (See, *e.g.*, `http://tug.org/errors.html`)

As reviewing will be double-blind, the submitted version of the papers should not include the authors' names and affiliations. Furthermore, self-references that reveal the author's identity, *e.g.*,

"We previously showed [**?**] ..."

should be avoided. Instead, use citations such as

"Gusfield:97 Gusfield:97 previously showed ... "

Any preliminary non-archival versions of submitted papers should be listed in the submission form but not in the review version of the paper. TTO 2019 reviewers are generally aware that authors may present preliminary versions of their work in other venues, but will not be provided the list of previous presentations from the submission form.

Please do not use anonymous citations and do not include when submitting your papers. Papers that do not conform to these requirements may be rejected without review.

References: Gather the full set of references together under the heading **References**. Arrange the references alphabetically by first author, rather than by order of occurrence in the text. By using a .bib file, as in this template, this will be automatically handled for you. See the `\bibliography` commands near the end for more.

Provide as complete a citation as possible, using a consistent format, such as the one for *Computational Linguistics* or the one in the *Publication Manual of the American Psychological Association* [**?**]. Use of full names for authors rather than initials is preferred. A list of abbreviations for common computer science journals can be found in the ACM *Computing Reviews* [**?**].

The LaTeX and BibTeX style files provided roughly fit the American Psychological Association format, allowing regular citations, short citations and multiple citations as described above.

- Example citing an arxiv paper: [**?**].
- Example article in journal citation: [**?**].
- Example article in proceedings, with location: [**?**].
- Example article in proceedings, without location: [**?**].

See corresponding .bib file for further details.

Submissions should accurately reference prior and related work, including code and data. If a piece of prior work appeared in multiple venues, the version that appeared in a refereed, archival venue should be referenced. If multiple versions of a piece of prior work exist, the one used by the authors should be referenced. Authors should not rely on automated citation indices to provide accurate references for prior and related work.

3.8 Footnotes

Footnotes: Put footnotes at the bottom of the page and use 9 point font. They may be numbered or referred to by asterisks or other symbols.[1] Footnotes should be separated from the text by a line.[2]

[1] This is how a footnote should appear.

[2] Note the line separating the footnotes from the text.

3.9 Graphics

Illustrations: Place figures, tables, and photographs in the paper near where they are first discussed, rather than at the end, if possible. Wide illustrations may run across both columns. Color illustrations are discouraged, unless you have verified that they will be understandable when printed in black ink.

Captions: Provide a caption for every illustration; number each one sequentially in the form: "Figure 1. Caption of the Figure." "Table 1. Caption of the Table." Type the captions of the figures and tables below the body, using 10 point text. Captions should be placed below illustrations. Captions that are one line are centered (see Table 1). Captions longer than one line are left-aligned (see Table 2). Do not overwrite the default caption sizes. The tto2019.sty file is compatible with the caption and subcaption packages; do not add optional arguments.

3.10 Accessibility

In an effort to accommodate people who are color-blind (as well as those printing to paper), grayscale readability for all accepted papers will be encouraged. Color is not forbidden, but authors should ensure that tables and figures do not rely solely on color to convey critical distinctions. A simple criterion: All curves and points in your figures should be clearly distinguishable without color.

4 Translation of non-English Terms

It is also advised to supplement non-English characters and terms with appropriate transliterations and/or translations since not all readers understand all such characters and terms. Inline transliteration or translation can be represented in the order of: original-form transliteration "translation".

Acknowledgments

The acknowledgments should go immediately before the references. Do not number the acknowledgments section. Do not include this section when submitting your paper for review.

Preparing References:

Include your own bib file like this: `\bibliographystyle{acl_natbib}` `\bibliography{acl2019}` where `acl2019` corresponds to a acl2019.bib file.